Final Exam Review — Comprehensive (Modules 01–15 plus Module 16 capstone synthesis)

BIOL-1

Name: _____ **Date:** _____

Learning Objectives

By the end of this review, you will be able to:

1. Connect the major ideas across all 16 modules — from the chemistry of life through evolution and ecology.
2. Trace mechanism chains that link molecular events to organismal, population, and ecosystem outcomes.
3. Practice free-response reasoning without relying on memorized wording.
4. Identify which concepts need more review before the final exam.

Purpose

This review pulls together ideas from every unit of BIOL-1. It is **not** a copy of the official final exam; wording and topics are reordered and rephrased so you practice **concepts**, not one memorized set of sentences.

How to use this review

1. Do the **written** items **closed-book** first, then reopen notes only where you were stuck.
2. For **short answers**, write in full sentences. **Definitions** should use **your** words — then compare to the key terms list.
3. Ask your instructor for the **official exam** instructions, timing, and point breakdown — those rules override anything informal here.

Unit 1: The Chemistry of Life (Modules 01–06)

Focus ideas

- **Module 01:** Life is recognized by shared properties (cells, energy use, response, reproduction, regulation, evolution). Scientific claims are tested with observations, variables, and repeatable reasoning.
- **Module 02:** Atoms form elements; covalent, ionic, and hydrogen bonds create different molecular properties. Water polarity explains cohesion, temperature buffering, and dissolving. pH tracks hydrogen ion concentration.
- **Module 03:** Carbon forms diverse skeletons. Dehydration synthesis builds polymers; hydrolysis breaks them. Carbohydrates, lipids, proteins, and nucleic acids each serve distinct functions.
- **Module 04:** Cell theory — cells are the basic units of life. Prokaryotic vs eukaryotic organization. Organelles divide cellular labor.
- **Module 05:** Phospholipid bilayers create selective permeability. Passive transport (diffusion, osmosis) vs active transport. Tonicity predicts water movement.
- **Module 06:** ATP couples energy-releasing and energy-requiring reactions. Enzymes lower activation energy. Photosynthesis builds sugar; respiration harvests usable energy.

Written practice (free response)

1. In **3–4** sentences, trace the path from a carbon atom in a CO_2 molecule in the air to a carbon atom in a muscle protein. Name the processes (photosynthesis, digestion, biosynthesis) and the types of molecules involved at each step.

2. A cell is placed in a hypertonic solution. In **2–3** sentences, describe what happens to the cell's volume and why. Then explain what would happen differently in a hypotonic solution.

3. An enzyme works best at 37°C. If you heat it to 80°C, activity drops to near zero. In **2–3** sentences, explain what happened to the enzyme and why it cannot simply keep working.

Quick check — write the letter (A–D) *optional*

A. Which type of bond is responsible for water's cohesion (surface tension)?
A) Covalent bonds within the water molecule
B) Hydrogen bonds between water molecules
C) Ionic bonds between water and salt
D) Peptide bonds in proteins

B. Which macromolecule stores genetic information?
A) Carbohydrate
B) Lipid
C) Protein
D) Nucleic acid

Unit 2: Genetics (Modules 07–11)

Focus ideas

- **Module 07:** DNA structure and base-pairing. Replication is semiconservative. Transcription produces RNA; translation builds proteins using codons. Mutations alter protein outcomes.
- **Module 08:** Mitosis produces identical daughter cells for growth/repair. Meiosis produces haploid gametes. Crossing over and independent assortment create variation. Nondisjunction changes chromosome number.
- **Module 09:** Alleles are gene variants. Mendelian inheritance predicts patterns. Punnett squares model probability. Incomplete dominance, codominance, polygenic traits, and environment complicate simple patterns.
- **Module 10:** Epigenetics — genes are switched on/off without changing the DNA sequence. DNA methylation and histone acetylation regulate expression. X-inactivation and environmental influences.
- **Module 11:** Genomics and biotechnology — restriction enzymes, PCR, gel electrophoresis, CRISPR-Cas9, and genetically modified organisms.

Written practice (free response)

4. Fill in the Central Dogma:

DNA → ___ → ___

5. A DNA template strand reads 3′-TACGCAATT-5′. Write the mRNA sequence and use a codon table to determine the amino acids.

- mRNA:
- Amino acids:

6. Cross two heterozygous tall plants (Tt × Tt). What is the expected phenotypic ratio?

7. In **2–3** sentences, explain why individuals do not evolve but populations do. Use the words **allele frequency** in your answer.

8. All cells have the same DNA. In **2–3** sentences, explain how a muscle cell and a brain cell become different, referencing epigenetic mechanisms.

Quick check — write the letter (A–D) *optional*

A. Which process produces haploid gametes?
A) Mitosis
B) Meiosis
C) Binary fission
D) DNA replication

B. A horse and a donkey produce a mule, but the mule is typically sterile. This outcome is:
A) A prezygotic barrier
B) A postzygotic barrier (reduced hybrid fertility)
C) Stabilizing selection
D) The founder effect

Unit 3: Evolution & Ecology (Modules 12–16)

Focus ideas

- **Module 12:** Darwin's "descent with modification." Natural selection requires variation, overproduction, and differential reproductive success. Evidence from fossils, biogeography, comparative anatomy, and molecular comparisons.
- **Module 13:** Microevolution = change in allele frequencies. Mutation, gene flow, drift, and selection can change alleles; nonrandom mating alone can change genotype proportions without changing allele frequencies. Compare bottleneck and founder effects. Use Hardy–Weinberg expectations under stated assumptions as a null model; a mismatch alone does not identify its cause.
- **Module 14:** Biological species concept. Prezygotic vs postzygotic barriers. Allopatric vs sympatric speciation.
- **Module 15:** Exponential (J) vs logistic (S) growth; carrying capacity (K). Density-dependent vs density-independent regulation. ~10% energy transfer per trophic level. Energy flows; nutrients cycle.
- **Module 16:** Capstone — biological explanations become stronger when they connect molecules, cells, organisms, populations, and ecosystems. Mechanism chains, feedback loops, constraints, and tradeoffs.

Written practice (free response)

9. In **2–3** sentences, define **evolutionary fitness** the way a biologist uses the term (not "strongest" or "survives longest").

10. A hurricane randomly kills 90% of a moth population; survivors repopulate. In **2–3** sentences, say whether this is best viewed as selection or drift, and name the specific drift label if it applies.

11. For **each** of directional, stabilizing, and disruptive selection, give **one** short phrase: what the environment favors.

12. Contrast **density-dependent** and **density-independent** limits in **2–3** sentences, with one example each.

13. With ~10% transfer per trophic level, explain in **2–3** sentences why supporting the same number of people with tuna (long chain) generally requires more primary production than eating closer to plants.

Quick check — write the letter (A–D) *optional*

A. A few lizards are accidentally transported to a new island; the new population only reflects the alleles the founders carried. This is:
A) The founder effect (genetic drift)
B) The bottleneck effect
C) Convergent evolution
D) Stabilizing selection

B. In ecosystems, chemical elements (nitrogen, carbon) are cycled, while energy is:
A) Cycled the same way as nitrogen
B) Lost as heat along the way; not recycled
C) Only stored in apex predators
D) Only found in the atmosphere

Cross-Course Synthesis

14. A drought changes plant growth in a local ecosystem. Some plant individuals carry a genetic variant that changes water-use efficiency. Trace a **six-step causal chain** from the genetic variant to an ecosystem-level effect.

15. Choose **two** course modules from different units (e.g., Module 06 and Module 15) and explain how each helps you reason about the drought scenario above.

Module	Concept used	How it supports your explanation

16. Where could **natural selection** act in the drought scenario? Identify the trait, the environment, and the fitness difference in **2–3** sentences.

17. Where could **feedback** occur in the drought scenario? Describe one reinforcing or balancing feedback loop in **2–3** sentences.

Word Bank Sprint

Use each term once. Practice meaning in context.

Word bank: *allele frequencies, allopatric, ATP, bilayer, denaturation, density-dependent, descent with modification, enzyme, homologous, natural selection, osmosis, postzygotic, stabilizing selection*

1. The cellular energy-coupling molecule is _____.

2. An _____ lowers activation energy without being consumed.

3. Excess heat causes _____ — loss of shape that disrupts enzyme function.

4. The phospholipid _____ forms the flexible boundary of a cell.

5. Water moving across a semipermeable membrane toward higher solute concentration is called _____.

6. In formal terms, microevolution means a change in _____ from one generation to the next.

7. Darwin's phrase for evolution: species share ancestry and accrue change over time — _____ with modification.

8. Forelimb bones in different tetrapods that share structure from a common ancestor are called _____ structures.

9. A birth-weight curve favoring the middle value is an example of _____.

10. A reproductive barrier that works after a zygote forms, such as hybrid sterility, is a _____ barrier.

11. A limiting factor that hits harder as a population crowds is _____ regulation.

12. Speciation that begins with geographic separation is often _____ speciation.

13. The mechanism of evolution that requires variation, heritability, and differential reproductive success is _____.

Self-Check Before the Exam

I can…	✓
List the shared properties that define life	[ ]
Describe how atoms form bonds and how water's polarity matters	[ ]
Compare the four macromolecule classes and their functions	[ ]
Compare prokaryotic vs eukaryotic cells and match organelles to roles	[ ]
Explain diffusion, osmosis, and active transport across membranes	[ ]
Describe ATP, enzymes, and how photosynthesis and respiration are linked	[ ]
Outline DNA replication, transcription, and translation	[ ]
Compare mitosis and meiosis and sources of variation	[ ]
Use Punnett squares and recognize non-Mendelian patterns	[ ]
Explain epigenetic regulation (methylation, acetylation, X-inactivation)	[ ]
Describe PCR, gel electrophoresis, and CRISPR	[ ]
State natural selection as variation + overproduction + differential success	[ ]
Distinguish allele-frequency processes from nonrandom mating; contrast drift vs gene flow	[ ]
Classify prezygotic vs postzygotic barriers; allopatric vs sympatric speciation	[ ]
Contrast J vs S growth; K; density-dependent vs independent; energy flow vs nutrient cycling; 10% rule	[ ]
Trace mechanism chains across biological scales and identify feedback	[ ]

Two terms I still mix up:

One action before the exam (be specific):

Final exam review worksheet, Modules 01–15 plus Module 16 capstone synthesis. BIOL-1: General Biology (Pelican Bay), Fall 2026 — College of the Redwoods